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AN ACCELERATED ELIMINATION TECHNIQUE FOR THE SOLUTION OF PERTURBED HAMILTONIAN SYSTEMS

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Abstract. A new technique is developed for the formal solution of non-degenerate or degenerate Hamiltonian systems under periodic perturbation through continually accelerated elimination of the periodic terms. Special features of the method are the ability to eliminate both short-period and long-period variables simultaneously and the attainment of [formal] 'quadratic convergence' for non-degenerate systems and nearly quadratic convergence in degenerate cases. The technique utilizes Lie transforms and is based on an approach due to Kolmogorov and Arnol'd.

1. Background – Basic Terminology

The primary object of this paper is the description of an accelerated technique for the formal solution of perturbed Hamiltonian systems through the elimination of periodic terms. Basic theory, method of application, and estimates on the speed of elimination are given in Section 2 of the paper. The next section applies the technique to two examples: a simple, nondegenerate system for which a complete solution is obtained, and the Main Problem of Lunar Theory, representative of degenerate Hamiltonians, for which only secular terms in the transformed Hamiltonian are determined. Finally the adaptation of the method to maintain the unperturbed frequencies of the system is detailed, as well as the use of the method in problems of deep resonance, in the last section.

Necessary to appreciate the differences between the new and previous practical methods are certain fundamental characteristics of perturbation solution. These, some useful terminology describing techniques of solution, and the salient features of constructive proofs of Kolmogorov and Arnol'd (on which the new technique is based) are given in this first section.

1.1. Properties of perturbation solution

A perturbed Hamiltonian system is generally described by a Hamiltonian function of the form

$$H(x, X; \varepsilon) = H_0(x, X; \varepsilon) + \varepsilon^m H_1(x, X; \varepsilon) \tag{1.1.1}$$

in which x and X are respectively the [n-vector] canonical coordinates and momenta. H_0 and H_1 are typically both power series in the available small parameter ε, but H_0 differs from H_1 in being soluble in form; we can then say that H_1 produces a *per-*

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turbation of order m (in ε) on H_0. Traditionally one endeavors to solve this system through the determination of a formal canonical transformation $(x, X) \leftrightarrow (y, Y; \varepsilon)$:

$$\begin{aligned} x &= x(y, Y; \varepsilon) \qquad & X &= X(y, Y; \varepsilon) \\ |x - y| &= O(\varepsilon^m) \qquad & |X - Y| &= O(\varepsilon^m) \end{aligned} \tag{1.1.2}$$

such that the Hamiltonian expressed in terms of the new variables

$$H^*(y, Y; \varepsilon) = H(x(y, Y; \varepsilon), X(y, Y; \varepsilon); \varepsilon) \tag{1.1.3}$$

is soluble. Typically such determination consists in the repeated application of a single process transforming a Hamiltonian of the form (1.1.1) into one of the form

$$H^*(y, Y; \varepsilon) = H_0^*(y, Y; \varepsilon) + \varepsilon^{m^*} H_1^*(y, Y; \varepsilon) \tag{1.1.4}$$

(where, again, H_0^* is soluble in form). The procedure bringing a Hamiltonian (1.1.1) to one of the form (1.1.4) shall be called a *step*. For $m^* > m$, we will say that the perturbation has been *eliminated to order* m^* (or *through order* $m^* - 1$); if $m^* - m = 1$ we will call such elimination *linear*, while if $m^* - m = m$ the elimination shall be termed *quadratic*. While a sequence of such steps can be composed into a single canonical transformation, it is important to note that we focus our attention on the individual steps themselves – each typically terminated by an explicit reevaluation of the new Hamiltonian.

In celestial mechanics, for example, the 'soluble form' H_0 is commonly a function free of momenta. In the particular case that the x are angular, we shall then call H_0 the *secular* part of the Hamiltonian. [We follow Brouwer's (1959) convention of identifying such angular variables as the momenta.] Corresponding to the equations (1.1.1) and (1.1.4) above, each step then brings a Hamiltonian

$$H(x, X; \varepsilon) = H_0(x; \varepsilon) + \varepsilon^m H_1(x, X; \varepsilon)$$

into one of the form

$$H^*(y, Y; \varepsilon) = H_0^*(y; \varepsilon) + \varepsilon^{m^*} H_1(y, Y; \varepsilon),$$

and 'perturbation' corresponds to momenta-dependent terms in the Hamiltonian. The ultimate goal is to obtain the transformed Hamiltonian independent of the momenta altogether

$$H^* = H^*(y, -; \varepsilon);$$

then, since canonical transformations preserve the form of the Hamiltonian equations of motion (Wintner, 1941, Section 26), the solution of the new Hamiltonian

$$y_i = y_{0_i} \qquad Y_i = -\left(\frac{\partial H^*(y_0)}{\partial y_i}\right) t + Y_{0i}$$

and the explicit transformation equations (1.1.2) yield the solution of the original system.

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A general result of the nature of the transformation (1.1.2) is the introduction of infinite Taylor series in (1.1.3), giving rise to questions of convergence of the resulting series. The first method to be faster than linearly eliminating was devised by Kolmogorov (1954) – a method described by Arnol'd (1963) as quadratically *convergent*. But while Kolmogorov's method was convergent in the strictest mathematical sense, no such claim generally can be made of other methods. It is to make this distinction that the terminology 'elimination' has been introduced. In the case that the transformed Hamiltonian and the coordinate transformation *do* converge, Birkhoff (and Moser) characterize the original Hamiltonians as *integrable*. Another difficulty with infinite series is the fact that practically one must accept elimination of the perturbation only to a finite order rather than identically. In practice, the momentum-free part of the transformed Hamiltonian is then accepted as representing the transformed system (to that order) and so giving the desired solution (again, to that order). But implicit in this procedure is the assumption that the transformed Hamiltonian is convergent so that the approximation of the transformed Hamiltonian by its soluble part is a good one.

Unfortunately, Siegel has shown (see, for example, Moser, 1968, Section 3(a)) that integrable Hamiltonians are highly exceptional in the sense that, in the space of all Hamiltonians (under an appropriate topology), they have measure 0; thus the approximation of a transformed Hamiltonian by its soluble part becomes problematical. Although Poincaré (1893, VIII, Section 3) showed that such formal solutions are *asymptotically* representative of the actual solution (i.e., the error between the formal and actual solutions vanishes as the small parameter does), the error is estimated by a function exponential in the time and can only hold to a given accuracy over a finite interval of time – an interval becoming shorter as the size of the order of elimination decreases. While Poincaré's result in some sense justifies the common formal procedures, it must be recognized that convergence generally does not exist.

An at least contributory reason for this characteristic divergence of Hamiltonians can be seen in the case common in celestial mechanics in which they are periodically dependent on the momenta:

$$H(x, X; \varepsilon) = H_0(x; \varepsilon) + \sum_{k \neq 0} H_{1k}(x; \varepsilon)\, e^{ik \cdot X}.$$

Since partials of this function with respect to the momenta yield the total time derivatives of the coordinates, it is the latter functions, also periodic in the momenta, which must be integrated. But since there is associated with each of the momenta a frequency, ω_i, there will appear in the integration of the coordinate time derivatives denominators of the form $(k \cdot \omega)$. In the event the frequencies are rationally dependent, such denominators will vanish identically; to avoid such singularities, attention must be restricted to frequencies rationally independent. On the other hand, by taking $|k|$ sufficiently large, it is possible to make $(k \cdot \omega)$ arbitrarily small; for this reason it is common to refer to even the nonresonant case as 'shallowly resonant'. Fortunately,

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shallow resonance need not threaten the convergence of the solution. For H_1 analytic in a nonempty closed region, the coefficients of the Fourier expansion, H_{1k}, generally decrease exponentially with $|k|$, and if the frequencies satisfy the *irrationality condition* (Arnol'd, 1963, V, Section 3.2)

$$|k \cdot \omega| \geqslant K|k|^{-(n+1)}, \qquad K = K(\omega) > 0,$$

the integrated series will still converge. Thus it is typical to make this assumption in any theoretical discussion of the techniques for solution of perturbed Hamiltonian systems.

However useful this condition is for mathematical description, though, it is meaningless when applied to actual analytical problems. Not only is it impossible to verify for all k; any numerical values for the ω will necessarily introduce rational representations violating the condition from the start. Thus it is common to ignore the question altogether. But there is even an additional difficulty: due to the fact that the frequencies of the transformed system are generally dependent on the [new] coordinates, the assignment of different values to these coordinates will possibly give frequencies violating the condition. This, in fact, is characteristic of integrable systems – the transformed Hamiltonian can be assigned arbitrary values of the coordinates without violating their convergence or that of the transformation equations; nonintegrable systems, on the other hand, might have assigned values of the coordinates resulting in resonance among the frequencies. (Recall, in this regard, the denseness of the rationals.)

Even rationally independent frequencies may be such that, for some moderate $|k|$, $|k \cdot \omega|$ may be small. (How 'moderate' is, in practice, a matter of judgement.) Such near resonance is observed in the mean motions of Jupiter and Saturn and arises from frequencies of approximately the same order. In certain cases, such 'deep resonance' may introduce a wholly new type of motion: libration instead of circulation.

It is also possible that the unperturbed frequencies may differ in order. Consider, for example, a Hamiltonian of the form

$$H(x, X; \varepsilon) = H_{00}(x; \varepsilon) + \varepsilon^q H_{01}(x; \varepsilon) + \varepsilon^m H_1(x, X; \varepsilon), \qquad m > q$$

in which certain coordinates are not present in H_{00} but appear instead for the first time in H_{01}; conjugate to such coordinates are canonical momenta termed *long period* (as opposed to the *short period* momenta associated with those coordinates present in H_{00}). In such a case linear combinations of the long period frequencies (which satisfy $k \neq 0$) will be of the order of ε^q. This characterizes *proper degeneracy* for Arnol'd (1963, I, Section 5). Although this appears to be just another way of generating small divisors, the fact that it is explicitly small (in ε) rather than implicitly so as in near resonance, makes it easier to handle than the latter case.

Even if the frequencies satisfy the irrationality condition, there is yet a second condition which must be fulfilled to prove the mathematical convergence of the infinite series: the available parameter must be not just 'small', but 'small enough'.

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While, given a constructive proof for convergence, it would be possible to estimate admissible values for this parameter – values themselves generally dependent on the coordinates, in the analytical expression describing a physical system, the small parameter will be predetermined by the system itself. Thus it is typical to ignore this matter, too, and merely to proceed with formal solution.

As a practical matter, then, imposition of conditions sufficient for mathematical proof of convergence must be supplanted by a merely formal technique in the solution of physical problems. But the application of a method firmly grounded mathematically appeals its own case against one more 'intuitive' in nature.

In summary, then, we see that an arbitrary Hamiltonian describing a physical system is almost certainly nonintegrable; consequently, the approximation of the transformed Hamiltonian by its integrable part in the common method of solution of Hamiltonian systems is not generally a good one (although such approximation may still give results valid over a finite time interval, by Poincaré's result). The frequencies are generally determined by the values assigned to the coordinates, and one suspects that a judicious choice of these values might allow the transformed series to converge. This, in fact, turns out to be the case and is fundamental to the proofs of Kolmogorov and Arnol'd demonstrating the convergence of their transformation.

Yet their proofs require the introduction of a new technique yielding quadratic convergence, realized only exceptionally in previous methods. Uniqueness of solution of perturbed differential equations with periodic perturbation then justifies the use of the classical methods. An important byproduct of these techniques is the accelerated elimination in comparison with classical procedures. It is the fundamental method of the Russian authors which is applied to obtain a practical method for faster-than-linear elimination of [periodic] momenta in Hamiltonian systems in this paper. A brief description of the Russians' methods in the perspective of the classical ones is useful to appreciate their new concept.

1.2. Review of perturbation methods

Fundamental to all these methods is the fact that, given any function of $2n$ variables, W, with nonvanishing Hessian (to ensure invertibility), the transformation $(x, X) \leftrightarrow (y, Y)$ given by

$$x_i = \frac{\partial W(y, X)}{\partial X_i} \qquad Y_i = \frac{\partial W(y, X)}{\partial y_i}$$

is canonical; W is called the *generating function* of the transformation. That the original coordinates and transformed momenta are given as functions of the new coordinates and old momenta is important, for since it is the explicit solution of the transformed Hamiltonian in terms of totally new variables that is known, it is desired to have

$$x = x(y, Y; \varepsilon) \qquad X = X(y, Y; \varepsilon).$$

To match the constants of integration in the transformed system to the original

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variables' initial conditions, it is also desirable to be able to express the new variables in terms of the old:

$$y = y(x, X; \varepsilon) \qquad Y = Y(x, X; \varepsilon).$$

Since the transformation equations are given in terms of mixed (old and new) variables, it is generally necessary after the elimination has been effected to solve for the variables only implicit in the transformation. While there is an algorithm (the Lagrange Inversion Formula) available for such explicit inversion, it introduces considerable complexity into the process of solution of the Hamiltonian system for even small orders of elimination. As will be seen presently, this is a major difficulty in application of the techniques of Kolmogorov and Arnol'd.

von Zeipel's Method. Typical – perhaps archetypical – of classical techniques is one generally referred to in celestial mechanics as the *von Zeipel* (or *Poincaré-von Zeipel*) *method.* Given a starting Hamiltonian

$$H(x, X; \varepsilon) = H_0(x) + \sum_{i \geq 1} \varepsilon^i H_i(x, X; \varepsilon), \tag{1.2.1}$$

it performs a canonical transformation through a generating function of the form

$$W = \sum_{i=1}^{n} y_i X_i + S(y, X), \qquad S = \sum_{j \geq 1} \varepsilon^j S_j(y, X)$$

yielding

$$\begin{aligned} x_i &= y_i + \sum_{j \geq 1} \varepsilon^j \frac{\partial S_j}{\partial X_i} \\ Y_i &= X_i + \sum_{j \geq 1} \varepsilon^j \frac{\partial S_j}{\partial y_i}. \end{aligned} \tag{1.2.2}$$

These transformations are then substituted into the H_i of (1.2.1), Taylor expansions performed, and like powers of ε collected to yield the transformed Hamiltonian

$$H_0\left(y + \frac{\partial S}{\partial X}\right) + \sum_{i \geq 1} \varepsilon^i H_i\left(y + \frac{\partial S}{\partial X}, X\right) \equiv \sum_{i \geq 0} \varepsilon^i H_i^*\left(y, X + \frac{\partial S}{\partial y}\right),$$

since the time-independent Hamiltonian is conserved. The orders on each side are equated, yielding generally an equation (Giacaglia, 1972, Equation 2.4.17) of the form

$$\sum_{i=1}^{n} \frac{\partial H_0(y)}{\partial y_i} \frac{\partial S_1(y, X)}{\partial X_i} + \varphi_j(y, X) + H_j(y, X) = H_j^*(y, X)$$

in which the φ_j are dependent on the H_k and S_k for $k < j$. Selection of the S_j give some freedom in determining the form of the H_j^*. Typically, $\varphi_j + H_j$ is separated into parts periodically dependent on, and independent of, the momenta through averaging

$$\varphi_j + H_j = \overline{H}_j(y) + \psi_j(y, X),$$

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and the S_i chosen to eliminate the periodic momenta (terms containing which correspond to the 'perturbation')

$$\sum_{j=1}^{n} \frac{\partial H_0}{\partial y_j} \frac{\partial S_i}{\partial X_j} + \psi_i(y, X) = 0, \tag{1.2.3}$$

yielding periodic S_i and thus, through (1.2.2), periodic transformations. In the case of proper degeneracy it is necessary to eliminate first the short period variables with one transformation, and then the long period variables with a subsequent one.

Important to the present discussion are two points. In the first place, the basic method involves the determination of a single transformation – at least two if the system is degenerate. Secondly, the *steps* (in the sense defined for this paper) are well-defined: at each order is involved the solution of the partial differential equation (1.2.3) and the determination of that order's contribution to the transformed Hamiltonian, H_i^*. Furthermore, the partial differential equation itself will generally depend on all the S_j for $j < i$, and, in particular, on S_{i-1}; thus the method is necessarily *linearily* eliminating of the momenta.

It should be mentioned again that practical application of this method involves the inversion of the transformation equations, expressed in terms of the mixed variables, to get the final solution – an inversion which must be performed at least twice in the degenerate case.

Kolmogorov's Method. Kolmogorov (Barrar, 1970) starts with a Hamiltonian of the form

$$\begin{aligned} H = H_0 + \sum \lambda_i x_i + \varepsilon(A(x) + \sum B_i(X) x_i) + \\ + \sum \sum C_{ij}(X; \varepsilon) x_i x_j + D(x, X; \varepsilon) \end{aligned}$$

(D of at least third degree in the x_i) – corresponding to the analytic expansion of an arbitrary Hamiltonian about, say, constant solutions to the unperturbed Hamiltonian. But his stratagem is quite different from that in the Poincaré-von Zeipel method; rather than eliminating identically the momenta, he seeks merely to eliminate momentum-dependent terms independent of, and linear in, the coordinates – the perturbations for Kolmogorov – and to obtain a final Hamiltonian of the form

$$H^*(y, Y; \varepsilon) = H_0 + \sum \lambda_i y_i + \sum \sum C_{ij}^*(Y; \varepsilon) y_i y_j + D^*(y, Y; \varepsilon)$$

admitting of the particular solution

$$y_i = 0 \qquad Y_i = \lambda_i t + Y_{0i}.$$

To effect this elimination, he performs a canonical transformation determined by the generating function

$$W(y, X; \varepsilon) = \sum (y_i + \xi_i) X_i + U(X; \varepsilon) + \sum V_j(X; \varepsilon) y_j, \tag{1.2.4}$$

in which ξ_i, V_i, and U are 'small', yielding

$$\begin{aligned} x_i &= (y_i + \xi_i) + \frac{\partial U}{\partial X_i} + \sum y_j \frac{\partial V_j}{\partial X_i} \\ Y_i &= X_i + V_i. \end{aligned} \tag{1.2.5}$$

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The transformed Hamiltonian then assumes the form

$$H(y, X; \varepsilon) = (H_0 + \sum \lambda_i \xi_i) + \sum \lambda_i y_i + A^*(X; \varepsilon) + \sum B_i^*(X; \varepsilon) y_i + \\ + A^{(1)}(X; \varepsilon) + \sum B_i^{(1)}(X; \varepsilon) y_i + \sum \sum C_{ij}^{(1)}(X; \varepsilon) y_i y_1 + \\ + D^{(1)}(y, X; \varepsilon),$$

in which

$$A^* = \sum \lambda_i \frac{\partial U}{\partial X_i} + A$$

$$B_i^* = \sum_j \lambda_j \frac{\partial V_i}{\partial X_j} + \sum_j C_{ij}\left(\xi_j + \frac{\partial U}{\partial X_j}\right) + B_i$$

$$A^{(1)} = \sum_i \sum_j C_{ij}\left(\xi_i + \frac{\partial U}{\partial X_i}\right)\left(\xi_j + \frac{\partial U}{\partial X_j}\right) + D\left(\xi_i + \frac{\partial U}{\partial X_i}, X\right)$$

$$B_i^{(1)} = \sum_k \sum_j C_{kj}\left(\xi_k + \frac{\partial U}{\partial X_k}\right) \frac{\partial V_i}{\partial X_j} + \sum_k \frac{\partial D}{\partial x_k}\left(\xi_i + \frac{\partial U}{\partial X_i}, X\right)\left(\delta_{ik} + \frac{\partial V_i}{\partial X_k}\right).$$

Note that the 'small' quantities occur in A^* and B_i^* at most to first order, while in $A^{(1)}$ and $B^{(1)}$ at least to second. ξ_i, V_i, and U are then determined to eliminate A^* and B_i^* identically, leaving the new Hamiltonian with 'perturbing' terms of second order and of exactly the same form as the original one. At this point, a totally *new* transformation defined by the generating function (1.2.4) but with new ξ_i^*, V_i^*, and U^* can be determined so that the subsequently transformed Hamiltonian will contain terms $A^{(2)}$ and $B_i^{(2)}$ of second order in the small quantities, themselves of second order in $A^{(1)}$ and $B_i^{(1)}$ – i.e., of fourth order in the *original* small quantities. Iteration of this technique effects *quadratic* elimination of the desired perturbing terms.

For Kolmogorov, then, a step consists of the determination of a totally *new* transformation applied to the Hamiltonian resulting from the immediately preceding transformation; the von Zeipel method, recall, determines serially successive orders of the *same* transformation (for nondegenerate systems) and its new Hamiltonian. It is the iterative nature of Kolmogorov's method which gives rise to the quadratic elimination – and ultimately *convergence* – reminiscent of Newton's method for obtaining roots.

Inversion, however, becomes a major obstacle to practical application of Kolmogorov's method. For each step a new transformation expressed in mixed variables must be determined. Performing the inversions to the potentially high orders to which elimination can be realized with the quadratic elimination is obviously unappealing.

One other important feature peculiar to Kolmogorov's approach is the nature of his solution. Notice that he starts with an unperturbed Hamiltonian $\sum \lambda_i x_i$ with *constant* frequencies. To ensure convergence, he imposes the irrationality condition on these frequencies. Having available a set satisfying this condition, he then systematically eliminates any alteration from these frequencies by determining the ξ_i to eliminate any constants resulting from the products in the term B_i^*; from the trans-

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formation equations (1.2.5), these ξ_i have the effect of shifting the origin of his new, constant, coordinates from the original – in some sense to alter the initial conditions on the system's coordinates to determine new frequencies of the perturbed system to match those of the original unperturbed system. Contrast this with the classical method, in which the frequencies are allowed to change as a result of continual addition of purely coordinate-dependent terms – to *determine* the new frequencies resulting from perturbation – and forcing variations from the original coordinates to be purely periodic in nature. While Kolmogorov is somewhat more flexible in the form of his finally desired Hamiltonian, still allowing the momenta to remain, he must accept only the particular final solution $z=0$, as opposed to the general constants allowed by the classical approach. One is reminded again, however, that maintenance of the irrationality condition so important to the mathematical proof of convergence is too fastidious in practical application.

Arnol'd's Method. Arnol'd's (1963) approach is basically the same as that of Kolmogorov, though it was designed originally for application to degenerate systems

$$\begin{aligned} H(x, X; \varepsilon) &= H_0(x; \varepsilon) + \varepsilon^m H_1(x; X; \varepsilon) = \\ &= (H_{00}(x) + \varepsilon H_{01}(x; \varepsilon)) + \varepsilon^m H_1(x, X; \varepsilon) \end{aligned} \tag{1.2.6}$$

in which all canonical coordinates appear in H_0, though some may be absent from the part H_{00}. H can be arbitrary in form like the von Zeipel method (and as distinct from Kolmogorov's analytic expansion); also classical is the regard of *all* momentum-dependent terms as perturbations. Unlike the von Zeipel method, however, elimination of both short- and long-period variables is carried out simultaneously at each step, instead of through the use of separate transformations.

The key to his method lies in the recognition that the relevant small parameter is not the original ε but (in Equation (1.2.6)) ε^m – the order of magnitude of the present perturbation – in contrast with the von Zeipel method, in which all eliminations are done in single powers of the original small parameter, regardless of the order to which the perturbation has been eliminated. A generating function is used to effect the canonical transformation, though, like Kolmogorov, the magnitude of this function (and so the variation from the original set of coordinates) is not predetermined but only 'small'. Although only the lowest order perturbation (ε^m) is eliminated at each step, the remaining perturbation is of the *square* of this in magnitude; even if the system is degenerate and there is a divisor of order ε, elimination is still effected since $m \geqslant 2$; indeed, for $m=2$, perturbation is eliminated to order $2^2-1=3$. This, in turn, becomes the new parameter, and a subsequent transformation will be generated by a function in whose denominator ε will again appear only to the first power, effecting an elimination to ε^5.

The mechanism of this approach can be seen explicitly for the above Hamiltonian (1.2.6); for purposes of simplicity, we shall assume H_2 to be purely periodic in X, thus avoiding for the nonce questions of alteration of the frequencies. As suggested by Arnol'd, the generating function S will be arbitrary, though in some sense 'small'.

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Expanding H in a Taylor series (in S) and collecting terms,

$$H = H_0 + \varepsilon H_1 + \left(\sum \left(\frac{\partial H_0}{\partial y_i} + \varepsilon \frac{\partial H_1}{\partial y_i}\right) \frac{\partial S}{\partial X_i} + \varepsilon^2 H_2\right) +$$

$$+ \tfrac{1}{2} \sum \left(\frac{\partial^2 H_0}{\partial y_i^2} + \varepsilon \frac{\partial^2 H_1}{\partial y_i^2} + \varepsilon^2 \frac{\partial^2 H_2}{\partial y_i^2}\right)\left(\frac{\partial S}{\partial X_i}\right)^2 +$$

$$+ \varepsilon^2 \sum \frac{\partial H_2}{\partial y_i} \frac{\partial S}{\partial X_i} + \frac{1}{6}\left(\frac{\partial^3 H_0}{\partial y_i^3} + \varepsilon \frac{\partial^3 H_1}{\partial y_i^3} + \varepsilon^2 \frac{\partial^2 H_2}{\partial y_i^3}\right)\left(\frac{\partial S}{\partial X_i}\right)^3 + \cdots.$$

Eliminating the first term in parentheses above

$$\sum \left(\frac{\partial H_0}{\partial y_i} + \varepsilon \frac{\partial H_1}{\partial y_i}\right) \frac{\partial S}{\partial X_i} = -\varepsilon^2 H_2 \tag{1.2.7}$$

then determines S, seen to be at least of order ε (so indeed 'small', as hypothesized). The remaining X-dependent part of the Hamiltonian is then easily seen to be of order ε^3, with the exception of the term

$$\sum \frac{\partial^2 H_0}{\partial y_i^2} \left(\frac{\partial S}{\partial X_i}\right)^2. \tag{1.2.8}$$

But, in fact, this is also; for from (1.2.7),

$$\sum \frac{\partial H_0}{\partial y_i} \frac{\partial S}{\partial X_i} = -\varepsilon^3 H_2 - \varepsilon \sum \frac{\partial H_1}{\partial y_i} \frac{\partial S}{\partial X_i},$$

so that, since S is already of order ε, (1.2.8) is of third order

As with all these methods, inversion is a very real problem – like in the Kolmogorov method, a problem compounded by the iterative nature of the technique.

Lie Transforms. The recurrent problem of inversion has been shown to be due to the use of a classical generation of the canonical transformations by a function dependent on both the original and transformed variables. Recently, however, Hori (1966) and Deprit (1969) have generated canonical transformations in terms of homogeneous variables, eliminating the final inversion. Both rely on a theory by Lie (1888, Section 12) who showed that the transformation $(x, X) \leftrightarrow (y, Y; \varepsilon)$

$$\begin{pmatrix} x \\ X \end{pmatrix} = \sum_{i \geq 0} \frac{\varepsilon^i}{i!} L_w^i \begin{pmatrix} y \\ Y \end{pmatrix},$$

where

$$L_w = \sum_{i=1}^{n} \frac{\partial W}{\partial Y_i} \frac{\partial}{\partial y_i} - \frac{\partial W}{\partial y_i} \frac{\partial}{\partial Y_i}$$

and

$$L_w^{i+1} = L_w L_w^i \qquad L_w^0 \equiv \text{identity},$$

is canonical. Both have developed algorithms which have gained wide acceptance. Their methods fundamentally recast the classical von Zeipel technique in terms of the

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new differential operator. Like this method, then, a transformation is determined to eliminate variables in single powers of the small parameter, i.e., linearly.

This paper, then, combines the iterative approach of Kolmogorov and Arnol'd with the Lie transform (after Deprit) to give a practical technique accelerating the elimination of momenta from a perturbed Hamiltonian. The Lie transforms make practical repeated transformations because of the relative ease of inversion. At the same time, due to their common occurrence in celestial mechanics, degenerate systems general in form are treated primarily in much the same way as Arnol'd's approach.

2. An Accelerated Elimination Technique for Hamiltonian Systems

Consider an arbitrary function $f(x, X; \kappa, \varepsilon)$ dependent on the two parameters κ and ε and analytic in the n-vector variables x and X at $\kappa=0$ and $\varepsilon=0$. Given any other C^∞ function, $W(x, X; \varepsilon)$, we define the *Poisson bracket of W with f*

$$L_W f \equiv \frac{\partial f}{\partial x_i}\frac{\partial W}{\partial X_i} - \frac{\partial f}{\partial X_i}\frac{\partial W}{\partial x_i}.$$

Then we can define recursively the nth iterate of L_W

$$L_W^n f \equiv L_W(L_W^{n-1} f),$$

in which

$$L_W^0 f \equiv f.$$

Now define the formal operator

$$\exp(\kappa L_W f) \equiv \sum_{i \geq 0} \frac{\kappa^i}{i!} L_W^i f$$

and consider the transformation $(x, X) \leftrightarrow (y, Y; \kappa, \varepsilon)$ defined by

$$\begin{pmatrix} x \\ X \end{pmatrix} = \exp\left(\kappa L_W \begin{pmatrix} y \\ Y \end{pmatrix}\right) = \begin{pmatrix} y \\ Y \end{pmatrix} + \frac{\kappa}{1!}\begin{pmatrix} \dfrac{\partial W}{\partial y} \\ -\dfrac{\partial W}{\partial y} \end{pmatrix} + \frac{\kappa^2}{2!}\begin{pmatrix} L_W \dfrac{\partial W}{\partial y} \\ -L_W \dfrac{\partial W}{\partial y} \end{pmatrix} + \cdots \quad \text{(I)}$$

in which now $W = W(y, Y; \varepsilon)$. (For ease of presentation, the properties of this transformation will be given as theorems; results fundamental to application will be identified with Roman numerals.)

THEOREM (Canonicity): *The transformation* (I) *is canonical.*

Proof: This proof follows similarly to those given by Hori (1966, Section 1) and Deprit (1969, Section 2).

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THEOREM (Transformation): *Under the transformation* (*I*)

$$f(\exp(\kappa L_W y), \exp(\kappa L_W Y); \varepsilon) = \exp(\kappa L_W f(y, Y; \varepsilon)). \tag{2.1.1}$$

(*This is called the* Lie transform of f under W.)

Proof: Again the proof is similar to the previous authors', making a Taylor expansion of f about $\kappa = 0$.

THEOREM (Inverse): *The inverse operator of* $\exp(\kappa L_W)$ *is* $\exp(\kappa L_{-W})$. (II)

Proof: Note first that, since $L_W^n W = 0$ for $n > 0$, (2.1.1) gives that

$$W(\exp(\kappa L_W y), \exp(\kappa L_W Y); \varepsilon) = W(y, Y; \varepsilon).$$

Now consider $f(x, X; \varepsilon)$, first under the transformation

$$\begin{pmatrix} x \\ X \end{pmatrix} = \exp\left(\kappa L_{W(y, Y; \varepsilon)} \begin{pmatrix} y \\ Y \end{pmatrix}\right)$$

yielding

$$f^*(y, Y; \kappa, \varepsilon) = \sum_{i \geq 0} \frac{\kappa^i}{i!} L^i_{W(y, Y; \varepsilon)} f(y, Y; \varepsilon)$$

and then under the second transformation

$$\begin{pmatrix} y \\ Y \end{pmatrix} = \exp\left(\kappa L_{-W(z, Z; \varepsilon)} \begin{pmatrix} z \\ Z \end{pmatrix}\right),$$

giving

$$f(y, Y; \varepsilon) = \sum_{j \geq 0} \frac{\kappa^j}{j!} L^j_{-W(z, Z; \varepsilon)} f(z, Z; \varepsilon).$$

Thus

$$\begin{aligned} f^{**}(z, Z; \kappa, \varepsilon) &= \sum_{i \geq 0} \sum_{j \geq 0} \frac{\kappa^{i+j}}{i!j!} L^i_{W(z, Z; \varepsilon)} L^j_{-W(z, Z; \varepsilon)} f(z, Z; \varepsilon) = \\ &= \sum_{k \geq 0} \frac{\kappa^k}{k!} \left(\left(\sum_{i=0}^{k} \binom{k}{i} L^i_{W(z, Z; \varepsilon)} L^{k-i}_{-W(z, Z; \varepsilon)} \right) f(z, Z; \varepsilon) \right) = \\ &= \sum_{k \geq 0} \frac{\kappa^k}{k!} (L_W + L_{-W})^k f(z, Z; \varepsilon) = f(z, Z; \varepsilon), \end{aligned}$$

i.e., the composition of the above transformations is just the identity.

(This compares directly with Equation (12) in Deprit's exposition (1969) of his Lie algorithm but applies only to the transformation of functions explicitly independent of the small parameter. Subsequent development to include functional dependence on the small parameter renders this result inapplicable to his final algorithm, although it does remain valid in the method presented here.)

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The following theorem on a particular form of function is that on which the new technique is based.

THEOREM: Consider a function of the form

$$g(x, X; \kappa, \varepsilon) = g_0(x; \varepsilon) + \kappa g_1(x, X; \varepsilon). \tag{2.1.2}$$

If $W(y, Y; \varepsilon)$ *satisfies the* determining equation

$$L_W g_0 + g_1 = \sum \left(\frac{\partial g_0}{\partial y_i} \frac{\partial W}{\partial Y_i} - \frac{\partial g_0}{\partial Y_i} \frac{\partial W}{\partial y_i} \right) + g_1 = 0, \qquad \text{(III)}$$

then

$$g(\exp(\kappa L_W y), \exp(\kappa L_W Y); \kappa, \varepsilon) = g_0(y; \varepsilon) + \sum_{i \geq 2} \kappa^i \frac{i-1}{i!} L_W^{i-1} \times$$
$$\times\, g_1(y, Y; \varepsilon); \qquad \text{(IV)}$$

i.e., *Y-dependent terms are eliminated to order* κ^2.

Proof: For a general generating function, W, (2.1.1) gives the Lie transform of (2.1.2):

$$g(\exp(\kappa L_W y), \exp(\kappa L_W Y); \kappa, \varepsilon) = \gamma_0 + \sum_{i \geq 0} \frac{\kappa^i}{i!} L_W^i (g_0 + \kappa g_1) =$$
$$= g_0 + \sum_{i \geq 1} \kappa^i \left(\frac{L_W^i g_0}{i!} + \frac{L_W^{i-1} g_1}{(i-1)!} \right) =$$
$$= g_0 + \sum_{i \geq 1} \frac{\kappa^i}{i!} L_W^{i-1} (L_W g_0 + i g_1).$$

But if W satisfies (III), the κ^1 term vanishes; and further, for $i \geqslant 2$,

$$L_W g_0 + i g_1 = (i-1) g_1,$$

from which the above result follows.

2.2. Application of the method

Consider a general Hamiltonian perturbation problem dependent on a single small parameter

$$\mathscr{H}(x, X; \varepsilon) = \sum_{i \geq 0} \varepsilon^i H_{0i}(x) + \sum_{i \geq m} \varepsilon^i H_{1i}(x, X) \equiv H_0(x; \varepsilon) +$$
$$+ H_1(x, X; \varepsilon) \tag{2.2.1}$$

in which

$$\frac{\partial H_0}{\partial x_i}(x; \varepsilon) \neq 0 \qquad i = 1, \ldots, n$$

(though $\partial H_{00}/\partial x_i$ may vanish).

Taking the lead of Kolmogorov and Arnol'd, we see that momentum-dependent

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terms, for us the perturbation, are of order ε^m, suggesting that the system be written

$$H(x, X; \kappa, \varepsilon) = \sum_{i \geq 0} \varepsilon^i H_{0i}(x) + \kappa \sum_{i \geq m} \varepsilon^{i-m} H_{1i}(x, X)$$

in which $\kappa = \varepsilon^m$. For X angular in nature and following the common practice of allowing the frequencies to change under perturbation – in some sense determining the frequencies as altered by the perturbation and disregarding the possible violation of the irrationality condition, we can 'average'

$$H_{1i}(x, X) = \bar{H}_{1i}(x) + \tilde{H}'_{1i}(x, X),$$

then define, for $i \geqslant m$,

$$H^*_{0i}(x) \equiv \bar{H}_{1i}(x), \qquad \tilde{H}_{1i}(x, X) \equiv \tilde{H}'_{1(i-m)}(x, X),$$

and $H^*_{0i} = H_{0i}$ for $i < m$, giving the Hamiltonian the desired form (2.1.2)

$$H(x, X; \kappa, \varepsilon) = \sum_{i \geq 0} \varepsilon^i H^*_{0i}(x) + \kappa \sum_{i \geq 0} \varepsilon^i \tilde{H}_{1i}(x, X).$$

The generating function is then determined using (III)

$$\sum_{k=1}^{n} \sum_{i \geq 0} \varepsilon^i \frac{\partial H^*_{0i}}{\partial y_k} \frac{\partial W}{\partial Y_k} + \sum_{i \geq 0} \varepsilon^i \tilde{H}_{1i} = 0 \tag{2.2.2}$$

and, with this $W = O(1)$ for nondegenerate Hamiltonians, under (I) the Hamiltonian

$$H^*(y, Y; \kappa, \varepsilon) = \sum_{i \geq 0} \varepsilon^i H^*_{0i}(y) + \sum_{j \geq 2} \kappa^j \frac{j-1}{j!} L_W^{j-1} \sum_{i \geq 0} \varepsilon^i \tilde{H}_{1i}(y, Y)$$

is obtained using (IV). At this point it can be recognized that $\kappa = \varepsilon^m$ 'in value', so the Hamiltonian can be written

$$\mathscr{H}^*(y, Y; \varepsilon) = H^*(y, Y; \varepsilon^m, \varepsilon) = \sum_{i \geq 0} \varepsilon^i H^*_{0i}(y) + \sum_{i \geq 2m} \varepsilon^i H^*_{1i}(y, Y).$$

But this is once again in the form (2.2.1) above so may be expressed

$$H^{**}(y, Y; \kappa^*, \varepsilon) = \sum_{i \geq 0} \varepsilon^i H^{**}_{0i}(y) + \kappa^* \sum_{i \geq 0} \varepsilon^i \tilde{H}^*_{1i}(y, Y),$$

in which now $\kappa^* = \varepsilon^{2m}$. Application of the transformation to this system eliminates the angular variables to order $\kappa^{*2} = \varepsilon^{4m}$ (for nondegenerate systems); subsequent transformations eliminate them or order ε^{8m}, ε^{16m}, *etc.*

At each stage the transformation (I) is made in terms of the homogeneous set of canonical variables (as opposed to the mixed set in the classical generating function approaches), giving $x = x(y, Y; \kappa, \varepsilon)$ and $X = X(y, Y; \kappa, \varepsilon)$ directly; if desired, the inverse transformation is easily found using (II) at each stage. Alternatively, a section of a recent paper by Henrard (1970, section 4) can likely be applied to give a single generating function yielding the transformation (and inverse) of the composition of a sequence of such transformations.

Crucial to the attainment of maximum elimination in the above is the *complete*

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solution of the determining equation (2.2.2) using the *entire* ε-expansion of H_0^*. For if the Hamiltonian is of the form

$$H = H_{01}(x; \varepsilon) + \varepsilon^{m'} H_{02}(x; \varepsilon) + \kappa \tilde{H}_1(x, X; \varepsilon), \qquad \varepsilon^{m'} > \kappa,$$

and W is determined neglecting higher-order terms in H_0

$$L_W H_{01} = -\tilde{H}_1,$$

the transformed Hamiltonian by (2.1.1) – (IV) is no longer applicable – will be of the form

$$H^* = H_{01} + \varepsilon^{m'} H_{02} + \sum_{i \geq 2} \kappa^i \frac{i-1}{i!} L_W^{i-1} \tilde{H}_1 + \varepsilon^{m'} \sum_{i \geq 1} \frac{\kappa^i}{i!} L_w^i H_{02}.$$

Since W generally depends on the momenta, this will introduce a Y-dependent term of magnitude $\kappa\varepsilon^{m'} > \kappa^2$. Though in certain situations this might be preferable to complete solution, the generally reduced elimination must be recognized.

Note that, due to the proposed form of starting Hamiltonian for each iteration, there will generally be unavoidable 'overlapping' of powers of ε. This apparently cumbersome procedure could have been avoided by considering the transformation of a starting Hamiltonian

$$H(x, X; \kappa, \varepsilon) = \sum_{i \geq 0} \varepsilon^i \sum_{j=0}^{m-1} \varepsilon^j H_{ij}(x, X)$$

but at the expense of simplicity of form in the functional transformation. Thus, unlike Deprit's algorithm, terms of like order in ε are not collected together automatically. It was with an eye toward manual application that the present scheme has been adopted.

It can be noted that the first step will coincide with that of a linear method.

Finally, as observed in the introductory remarks to this section, the starting Hamiltonian (2.2.1) may correspond to a degenerate one; the algorithm is identical in this case. As with the Arnol'd method, the elimination will generally be less than quadratic, though still accelerated, as shown by the following.

2.3. Elimination properties of the method

THEOREM (Perturbation Elimination): *Consider a Hamiltonian*

$$H(x, X; \kappa, \varepsilon) = H_0(x; \varepsilon) + \varepsilon^m H_1(x, X; \varepsilon)$$

in which $H_1 = O(1)$ *and*

$$\frac{\partial H_0}{\partial x_i}(x; \varepsilon) \geqslant 0(q)$$

for $q < m$ *and for all* $i = 1, \ldots, n$. *Allow this Hamiltonian to be subject to successive*

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transformations of the form (I) in which the generating functions satisfy (III) at each stage; then, after p steps, the transformed Hamiltonian $H^{(p)}$ is of the form

$$H^{(p)}(y, Y; \varepsilon) = H_0(y; \varepsilon) + \varepsilon^m H^{(p)}(y; \varepsilon) + \varepsilon^{2^p m - (2^p - 1)q} H_1^{(p)}(y, Y; \varepsilon) \tag{2.3.1}$$

where $H_1^{(p)} = O(1)$.

Proof: This follows immediately by induction and the observation that, since $W^{(p)} = O(\varepsilon^{-q})$ and $H^{(p)} \equiv O(1)$ for all p,

$$O(L_W^{i-1} H_1^{(p)}) = O(L_W^{i-1}) = O(\varepsilon^{(1-i)q}),$$

so that

$$\kappa^i L_W^{i-1} H_1^{(p)} = O(\varepsilon^{2^p m i - (2^p i - 1)q}).$$

Thus, by (IV), the lowest-order momentum-dependent term corresponds to $i=2$, from which the above estimate follows.

3. Examples Illustrating the Method

3.1. Complete solution through fifth order of a simple one-degree-of-freedom system

Since a simple, even artificial, example will often best illustrate the mechanism of a new technique, we shall apply the method to the one-degree-of-freedom system

$$\mathscr{H}(x, X; \varepsilon) = \omega x + \varepsilon(x + \eta) \sin X + \varepsilon^2 2x \sin X$$

corresponding to a perturbed simple harmonic oscillator.

Elimination. Noting that the perturbation is of order ε, we set the perturbation parameter $\kappa = \varepsilon$; thus

$$\begin{aligned} H(x, X; \kappa, \varepsilon) &= \omega x + \kappa((1 + 2\varepsilon)x + y) \sin X \equiv \\ &\equiv H_0(x) + \kappa H_1(x, X; \varepsilon). \end{aligned}$$

Making the formal transformation (I)

$$\begin{pmatrix} x \\ X \end{pmatrix} = \exp\left(\kappa L_W \begin{pmatrix} y \\ Y \end{pmatrix}\right) \tag{3.1.1}$$

W is found by solving the determining equation (III)

$$L_W H_0 = \omega \frac{\partial W}{\partial Y} = -((1 + 2\varepsilon) + y) \sin Y = -H_1$$

since H_1 is already purely periodic; thus

$$W = \frac{((1 + 2\varepsilon)y + \eta) \cos Y}{\omega}. \tag{3.1.2}$$

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Calculation of the explicit transformed Hamiltonian by (IV) requires the $L_W^i H_1$:

$$L_W H_1 = -\frac{(1+2\varepsilon)^2 y + (1+2\varepsilon)\eta}{\omega}$$

$$L_W^2 H_1 = \frac{(1+2\varepsilon)^3 y + (1+2\varepsilon)^2 \eta}{\omega^2} \sin Y.$$

$L_W H_1$, the second-order term in κ in the transformed Hamiltonian, is free of Y; this is generally the case for a single-degree-of-freedom oscillator with the present method. Thus the periodic part of the new function will be at least order $\kappa^3 = \varepsilon^3$, and the next stage will eliminate periodic terms to order $(\varepsilon^3)^2 = \varepsilon^6$, recommending expansion to $\kappa^5 = \varepsilon^5$ coefficients of the $L_W^i H_1$ through $i=4$:

$$L_W^3 H_1 = -\frac{(1+2\varepsilon)^4 y + (1+2\varepsilon)^3 \eta}{\omega^3}$$

$$L_W^4 H_1 = \frac{(1+2\varepsilon)^5 y + (1+2\varepsilon)^4 \eta}{\omega^4} \sin Y.$$

Thus the new transformed Hamiltonian is

$$\begin{aligned} H^*(y, Y; \kappa, \varepsilon) &= H_0(y) + \sum_{i \geq 2} \kappa^i \frac{i-1}{i!} L_W^{i-1} H_1 = \\ &= \omega y + \frac{\kappa^2}{2}\left(-\frac{(1+2\varepsilon)^2 y + (1+2\varepsilon)\eta}{\omega}\right) + \\ &\quad + \frac{\kappa^3}{3}\left(\frac{(1+2\varepsilon)^3 y + (1+2\varepsilon)^2 \eta}{\omega^2} \sin Y\right) + \\ &\quad + \frac{\kappa^4}{8}\left(-\frac{(1+2\varepsilon)^4 y + (1+2\varepsilon)^3 \eta}{\omega^3}\right) + \\ &\quad + \frac{\kappa^5}{30}\left(\frac{(1+2\varepsilon)^5 y + (1+2\varepsilon)^4 \eta}{\omega^4} \sin Y\right) + \kappa^6 \ldots . \end{aligned}$$

Replacing κ by ε, expanding the terms $(1+2\varepsilon)^i$ to ε^5, and collecting like powers of ε, we get

$$\begin{aligned} H^*(y, Y; \varepsilon, \varepsilon) &= \omega y + \varepsilon^2\left(-\frac{y}{2\omega}\right) + \varepsilon^3\left(-\frac{2}{\omega} y + \frac{y+\eta}{3\omega^2} \sin Y\right) + \\ &\quad + \varepsilon^4\left(-\frac{2}{\omega} y - \frac{y}{8\omega^2} + \frac{6y+4\eta}{3\omega^2} \sin Y\right) + \\ &\quad + \varepsilon^5\left(-\frac{y}{\omega^3} + \left(\frac{12y+4\eta}{3\omega^2} + \frac{(y+\eta)}{30\omega^4}\right) \sin Y\right) + \cdots . \end{aligned}$$

This terminates the first step.

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The second step starts with the observation that perturbation is of order ε^3, suggesting that we make the new perturbation parameter $\kappa^* = \varepsilon^3$:

$$H^*(y, Y; \kappa^*, \varepsilon) = \omega y - \varepsilon^2 \frac{y}{2\omega} - \varepsilon^3 \frac{2y}{\omega} - \varepsilon^4\left(\frac{2y}{\omega} + \frac{y}{8\omega^2}\right) - \varepsilon^5 \frac{y}{\omega^2} + \\ + \kappa^*\left(\frac{y+\eta}{3\omega^2} + \varepsilon \frac{6y + 4\eta}{3\omega^2} + \varepsilon^2\left(\frac{12y + 4\eta}{3\omega^2} + \right.\right. \\ \left.\left. + \frac{y+\eta}{30\omega^4}\right)\right) \sin Y \equiv H_0^*(y; \varepsilon) + \kappa^* \tilde{H}_1^*(y, Y; \varepsilon),$$

in which the *entire* secular part

$$H_0^*(y; \varepsilon) \equiv \omega^*(\varepsilon) y$$

must assume the role of the unperturbed Hamiltonian to effect maximum elimination. We now make a second transformation

$$\begin{pmatrix} y \\ Y \end{pmatrix} = \exp\left(\kappa^* L_V \begin{pmatrix} z \\ Z \end{pmatrix}\right), \tag{3.1.3}$$

where V is found from the determining equation (III)

$$L_V H_0^* = -\tilde{H}_1^*$$

or

$$\omega^* \frac{\partial V}{\partial Z} = -\tilde{H}_1^*,$$

giving

$$V = \frac{1}{\omega^*}\left(\frac{z+\eta}{3\omega^2} + \varepsilon \frac{6z + 4\eta}{3\omega^2} + \varepsilon^2\left(\frac{12z + 4\eta}{3\omega^2} + \frac{z+\eta}{30\omega^4}\right)\right) \sin Z \tag{3.1.4}$$

to eliminate angle-dependent terms to order $\kappa^{*2} = \varepsilon^6$. The new Hamiltonian (accurate through ε^5) is

$$H^{**}(z) = H_0^*(z; \varepsilon) = \omega^* z.$$

This concludes the second step in the elimination.

Inversion. To complete the solution, it is necessary to express (x, X) in terms of the solution (z, Z) of the above Hamiltonian

$$z = z_0 \qquad Z = \omega^* t + Z_0.$$

Since the solution is good through ε^5, all transformations must be carried to this order (or its equivalent in terms of κ and κ^*).

First we find, from (3.1.1)

$$x = \exp(\kappa L_W y) = \sum_{i \geq 0} \frac{\kappa^i}{i!} L_W^i y$$

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where $\kappa=\varepsilon$, so we must find the $L_W^l y$ through $l=5$ using (3.1.2):

$$L_W y = \frac{\partial W}{\partial Y} = -\frac{(1+2\varepsilon)y+\eta}{\omega}\sin Y$$

$$L_W^2 y = L_W(L_W y) = \frac{(1+2\varepsilon)^2 y + (1+2\varepsilon)\eta}{\omega^2}.$$

$$L_W^3 y = -\frac{(1+2\varepsilon)^3 y + (1+2\varepsilon)^2\eta}{\omega^3}\sin Y$$

$$L_W^4 y = \frac{(1+2\varepsilon)^4 y + (1+2\varepsilon)^3\eta}{\omega^4}$$

$$L_W^5 y = -\frac{(1+2\varepsilon)^5 y + (1+2\varepsilon)^4\eta}{\omega^5}\sin Y;$$

so that, since $\kappa=\varepsilon$,

$$x = y - \varepsilon\left(\frac{(1+2\varepsilon)y+\eta}{\omega}\sin Y\right) + \frac{\varepsilon^2}{2}\left(\left(\frac{1+2\varepsilon}{\omega}\right)\frac{(1+2\varepsilon)y+\eta}{\omega}\right) -$$
$$-\frac{\varepsilon^3}{6}\left(\left(\frac{1+2\varepsilon}{\omega}\right)^2\frac{(1+2\varepsilon)y+\eta}{\omega}\sin Y\right) +$$
$$+\frac{\varepsilon^4}{24}\left(\left(\frac{1+2\varepsilon}{\omega}\right)^3\left(\frac{(1+2\varepsilon)y+\eta}{\omega}\right)\right) -$$
$$-\frac{\varepsilon^5}{120}\left(\left(\frac{1+2\varepsilon}{\omega}\right)^4\frac{(1+2\varepsilon)y+\eta}{\omega}\sin Y\right) + \varepsilon^6 \ldots .$$

The next transformation (3.1.3)

$$y = \exp\left(\kappa^* L_V\begin{pmatrix} z \\ Z \end{pmatrix}\right)$$

will give $x(z, Z; \kappa^*(\varepsilon))$. To go to ε^5 it is required to go at most to κ^{*1} in the transformation $\exp(\kappa^* L_V x(y, Y; \varepsilon))$; such transformation will (to this order) affect only terms factored by powers of ε through the second. Making the transformation on these terms, replacing (y, Y) by (z, Z) in the balance, and collecting like powers of ε, finally results in

$$x = z - \left(\frac{z+\eta}{\omega}\sin Z\right) + \varepsilon^2\left(\frac{z+\eta}{2\omega^2} - \frac{2z}{\omega}\sin Z\right) +$$
$$+\varepsilon^3\left(\frac{2z+\eta}{\omega^2} - \frac{z+\eta}{\omega^2}\left(\frac{1}{3\omega^*} + \frac{1}{6\omega}\right)\sin Z\right) +$$
$$+\varepsilon^4\left(\frac{3(z+\eta)}{8\omega^4} + \frac{2z}{\omega^2} - \frac{3z+2\eta}{\omega^4}\sin Z\right) +$$
$$+\varepsilon^5\left(\frac{38z+29\eta}{12\omega^4} - \left(\frac{6z+2\eta}{\omega^3} + \frac{z+\eta}{24\omega^5}\right)\sin Z\right) + \varepsilon^6 \ldots$$

in which the fact that $\omega=\omega^*$ to order ε^2 has been used in the last two terms.

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An exactly similar procedure is carried out to express $X(z, Z; \varepsilon)$:

$$X = Z - \varepsilon\left(\frac{\cos Z}{\omega}\right) - \varepsilon^2\left(\frac{2\cos Z}{\omega} + \frac{\sin 2Z}{4\omega^2}\right) +$$
$$+ \varepsilon^3\left(\left(\frac{1}{12\omega^3} - \frac{1}{3\omega^2\omega^*}\right)\cos Z - \frac{\sin 2Z}{\omega^2} + \frac{\cos 3Z}{12\omega^3}\right) +$$
$$+ \varepsilon^4\left(-\frac{3\cos Z}{2\omega^3} - \left(\frac{1}{8\omega^4} + \frac{1}{\omega^2}\right)\sin 2Z + \frac{\cos 3Z}{2\omega^3} + \frac{\sin 4Z}{32\omega^4}\right) -$$
$$- \varepsilon^5\left(\left(\frac{1}{24\omega^5} - \frac{3}{\omega^3}\right)\cos Z - \frac{\sin 2Z}{\omega^4} + \left(\frac{1}{16\omega^5} + \frac{1}{\omega^3}\right)\cos 3Z +\right.$$
$$\left. + \frac{\sin 4Z}{4\omega^4} - \frac{\sin 5Z}{80\omega^5}\right) + \varepsilon^5 \ldots .$$

The latter solution in particular has been checked to this order by an independent perturbation method.

It must be emphasized that the elimination through ε^5 is a special case due to the particularly simple starting Hamiltonian; nevertheless, elimination would generally be effected through ε^3, in contrast with the elimination through only ε^2 given by two steps with a linearly eliminating method. (The generally accelerated elimination is perhaps more obvious in the second transformation above.)

3.2. Determination of Secular Terms through Fourth-Order in the Main Problem of Lunar Theory

Representative of degenerate Hamiltonians common in celestial mechanics (due to the use of the two-body problem as unperturbed Hamiltonian) is the main problem of lunar theory expressed in Delaunay variables (Hori, 1963, Equation (8))

$$\mathscr{H}(x, X; \varepsilon) = \frac{\mu}{2x_1^2} + \nu x_2 + \nu^2 \frac{a^2}{4}\left[\left(\frac{r}{a}\right)^2 + 3\left(\frac{r}{a}\right)^2 \cos(2f + 2X_2)\right]. \quad (3.2.1)$$

In the interests of demonstrating the elimination properties of the new method, the canonical transformations only will be carried out explicitly; final solutions could be obtained with the generating functions determined.

Dut to its degeneracy, the system would be expected to have angles eliminated only through order ν^4 after two iterations by (2.3.1); in addition we expect a generating function of the form

$$W(y, Y; \nu) = \frac{1}{\nu} W_{-1}(y, Y) + W_0(y, Y) + \nu W_1(y, Y) + \cdots . \quad (3.2.2)$$

(Note that there is no problem with ν in the denominator, since in all transformations it is multiplied by a power of $\kappa = \nu^2$.) Because the present method allows such divisors in the generating function, some advance planning is necessary to ascertain the order

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in ν to which W must be determined and the order in κ to which the Hamiltonian transformation must be taken to get the transformed function

$$H^*(y, Y; \kappa, \varepsilon) = H_0(y; \varepsilon) + \frac{\kappa^2}{2} L_W H_1(y, Y; \varepsilon) + \\ + \frac{\kappa^3}{3} L_W^2 H_1(y, Y; \varepsilon) + \cdots \tag{3.2.3}$$

accurate to order ν^4. The former is easy: since the lowest-order term involving W is the $\kappa^2 = \nu^4$ term, it suffices to know W to 0th order to go to the desired accuracy in the first transformation. The most straightforward way of determining the order necessary in the Hamiltonian transformation is to note that, since at this stage $\kappa = \nu^2$ and $W = O(\nu^{-1})$,

$$\kappa L_W^{i-1} H_1 = O(\nu^{i+1})$$

(just as in the proof of the elimination property above, since $H_1 \equiv O(1)$), so it is required to go to $i = 3$ – i.e., κ^3 – to get the transformed Hamiltonian terms in ν^4. Having determined this, we also observe that, since

$$L_{(\Sigma \nu^i W_i)} H_1 = \sum \nu^i L_{W_i} H_1,$$

then

$$H^*(y, Y; \kappa, \nu) = H_0(y; \nu) + \frac{\kappa^2}{2}\left(\frac{1}{\nu} L_{W_{-1}} H_1 + L_{W_0} H_1 + \cdots\right) + \\ + \frac{\kappa^3}{3}\left(\frac{1}{\nu^2} L_{W_{-1}} H_1 + \cdots\right) + O(\nu^5). \tag{3.2.4}$$

To maintain a closed-form solution, necessary integrals will be done with reference to the true anomaly, f, and the mean anomaly, u, as is common in celestial mechanics. In order to expedite such calculations, Hori immediately writes the ν^2 term in (3.2.1) in the form

$$\frac{a^2}{4}(1 + \tfrac{3}{2}e^2) + \frac{a^2}{4}(A + (B + \tfrac{15}{2}e^2)\cos 2X_2 + C \sin 2X_2),$$

in which (Hori's Equation (17))

$$A = -e^2 - 2e \cos u + \tfrac{1}{2}e^2 \cos 2u \\ B = -3e^2 - 6e \cos u + (3 - \tfrac{3}{2}e^2) \cos 2u \\ C = \eta(6e \sin u - 3 \sin 2u), \qquad \eta = (1 - e^2)^{1/2}$$

are periodic in $Y_1(u)$. Thus, finally, we have the form of the starting Hamiltonian on which we shall operate:

$$H = \left(\frac{\mu^2}{2x_1^2} + \nu x_2 + \nu^2 \frac{a^2}{4}(1 + \tfrac{3}{2}e^2)\right) + \\ + \nu^2 \frac{a^2}{4}(A + (B + \tfrac{15}{2}e^2)\cos 2X_2 + C \sin 2X_2) \equiv \\ \equiv H_0(x; \nu) + \kappa H_1(x, X),$$

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where $\kappa = \nu^2$. The determining Equation (III)

$$L_W H_0 = -H_1$$

for the first transformation can be obtained by writing

$$\omega_1 = \frac{\partial H_0}{\partial y_1} = -\frac{\mu^2}{y_1^3} + \nu^2 \frac{y_1^3}{4\mu^2}\left(10 - \frac{3y_2^2}{y_1^2}\right) \equiv \omega_{10} + \nu^2\omega_{12}$$

$$\omega_2 = \frac{\partial H_0}{\partial y_2} = \nu - \nu^2 \frac{3y_1^2 y_2}{4\mu^2} \equiv \nu\omega_{21} + \nu^2\omega_{22},$$

substituting (3.2.2) into this equation, yielding

$$\left(\omega_{10}\frac{\partial W_0}{\partial Y_1} + \omega_{21}\frac{\partial W_{-1}}{\partial Y_2}\right) + \nu\left(\omega_{10}\frac{\partial W_1}{\partial Y_1} + \omega_{21}\frac{\partial W_0}{\partial Y_2} + \omega_{22}\frac{\partial W_{-1}}{\partial Y_2}\right) + \\ + \nu^2 \ldots = -H_1,$$

and then recursively forcing terms factored by positive powers of ν to vanish identically; this procedure gives W to the order required:

$$W_{-1} = -\frac{15y_1^4}{16\mu^2} e^2 \sin 2Y_2$$

$$W_0 = \frac{y_1^7}{4\mu^4}\left({}^{(1)}A + {}^{(1)}B\cos 2Y_2 + {}^{(1)}C\sin 2Y_2\right) - \frac{45y_1^6 y_2 e^2}{64\mu^4}\sin 2Y_2$$

where the new functions (Hori's Equation (18))

$$\begin{aligned}
{}^{(1)}A &= (-2e + \tfrac{3}{4}e^3)\sin u + \tfrac{3}{4}e^2 \sin 2u - \tfrac{1}{12}e^3 \sin 3u \\
{}^{(1)}B &= (-\tfrac{15}{2}e + \tfrac{15}{4}e^3)\sin u + (\tfrac{3}{2} + \tfrac{3}{4}e^2)\sin 2u + (-\tfrac{1}{2}e + \tfrac{1}{4}e^3)\sin 3u \\
{}^{(1)}C &= \eta[-\tfrac{15}{4}e^2 - \tfrac{15}{2}e\cos u + (\tfrac{3}{2} + \tfrac{3}{2}e^2)\cos 2u - \tfrac{1}{2}e\cos 3u]
\end{aligned}$$

are periodic in Y_1. Such W brings the transformed Hamiltonian into the form (3.2.3), which would normally be the starting function for a subsequent transformation. In this case, however, we shall only calculate the secular terms produced by the first transformation.

To do so, we recall that such secular terms, due to the fact that W is purely periodic, can only result from the multiplications of trigonometric terms of like argument. By (3.2.4) then, we see that it will be necessary to have explicitly those terms from $L_{W_{-1}}H_1$ of argument $(0Y_1 + 2Y_2)$ to get secular terms to fourth order arising from the term $L^2_{W_{-1}}H_1$; none of the other terms in the actual transformation need be calculated. (Were it desired to go to higher order, of course, the complete general terms would have to be calculated explicitly.) Proceeding thusly, we obtain the expression

$$L_{W_{-1}}H_1 = \frac{225y_1^7 e^2\eta}{32\mu^4} + \frac{15}{32\mu}\left(y_1^7\eta e\frac{\partial A}{\partial e}\cos 2Y_2 + \right. \\ \left. + y_1^4(2y_1^3e^2 + y_1y_2^2e)\frac{\partial A}{\partial Y_1}\sin 2Y_2\right) + \cdots$$

containing the secular contribution and terms with relevant argument.

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Only secular terms of $L_{W_0}H_1$ need be found; these result in

$$(L_{W_0}H_1)_5 = \frac{y_1^{10}}{64\mu^6}(-98 + \tfrac{4167}{2}e^2 - \tfrac{12069}{8}e^4).$$

Finally, it happens that calculation of $L^2_{W_{-1}}H_1$ introduces no further secular terms (which would be of order ν^4 if present). Substituting the above results into (3.2.4) in which κ is replaced by ν^2 results in

$$\begin{aligned} H^*(y, Y; \nu^2, \nu) = {} & \left(\frac{\mu^2}{2y_1^2} + \nu y_2 + \nu^2 \frac{a^2}{4}(1 + \tfrac{3}{2}e^2)\right) + \nu^3 \frac{225 y_1^7 e^3 \eta}{64\mu^4} + \\ & + \nu^4 \frac{y_1^{10}}{128\mu^6}(-98 + \tfrac{4167}{2}e^2 - \tfrac{12069}{8}e^4) + \\ & + \nu^3(\text{purely periodic terms in } Y) + \cdots \equiv H_0^* + \nu^3 \tilde{H}_1^*. \end{aligned} \tag{3.2.5}$$

As was mentioned in the section detailing application of the method, the periodic terms (had they been calculated explicitly) could be removed by a subsequent transformation

$$\begin{pmatrix} Y \\ y \end{pmatrix} = \exp\left(\kappa^* L_V \begin{pmatrix} z \\ Z \end{pmatrix}\right)$$

in which $\kappa^* = \nu^3$, where V would satisfy

$$L_V H_0^* = -\tilde{H}_1^*.$$

By (2.3.1), this elimination would leave perturbing terms of order ν^5 in the transformed Hamiltonian to be removed subsequently by a transformation affecting terms of this order and higher. Thus (3.2.5) represents (upon elimination of the periodic terms by V above) the finally angle-eliminated Hamiltonian through ν^4. The result checks with Hori's result (1963, Equation (69)).

Note that after two transformations (generated respectively by W and V), angles have been eliminated through ν^4, in contrast with the minimum of three expected from a linearly eliminating technique. Due to the degeneracy of the system, however, a total of *six* separate steps is generally required linearly to eliminate respectively the short- and long-period variables in the von Zeipel method. Had the second transformation above been carried out explicitly, averaging terms in this second transformed Hamiltonian through ν^8 would have given secular terms invariant under a third transformation (of order ν^9 by (2.3.1)), while the linear elimination techniques would have required a total of twelve steps. Although the final solution (given by the explicit transformations generated by W, V, ...) will vary periodically from certain constant values, the arguments of these periodic terms will be linear in the time; thus it is important that the frequencies be known as accurately as possible. This becomes perhaps the strongest argument in favor of the new technique, since at any stage, secular terms eliminated at the next stage will be invariant under subsequent transformations to *twice* (or slightly less, accounting for the effects of degeneracy) the

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order of the present perturbation, instead of the single higher order given by linear methods; in some sense, the transformation has sufficient information to determine the new Hamiltonian to this order, while linear methods have only enough to go to the next higher order. The economy to be realized in going to even moderately high order in solution, especially in degenerate systems typical of celestial mechanics, is evident.

4. Special Cases

4.1. Maintenance of initial frequencies

Though generally one is interested in the determination of alteration of frequencies due to perturbation, in certain instances a certain set of frequencies might be desired to remain unaltered under a perturbation; in this case it would generally be necessary to alter the initial conditions of the problem so that the perturbed system would maintain its frequencies. By modifying the form of the generating function to make it perform more like Kolmogorov's (1954, though more general in form), this can be done.

For consider a generating function of the form

$$W = \xi \cdot Y + W(y, Y; \varepsilon);$$

then, under (I), the transformed Hamiltonian assumes the form

$$H^* = H_0 + \kappa\left(\sum \frac{\partial H_0}{\partial y_i}\left(\xi_i + \frac{\partial W}{\partial Y_i}\right) + \bar{H}_1 + \tilde{H}_1\right) + \kappa^2 \ldots .$$

The separate parts of W are then determined such that

$$\sum \frac{\partial H_0}{\partial y_i}\frac{\partial W}{\partial Y_i} = -\tilde{H}_1$$

$$\sum_i \xi_i \frac{\partial^2 H_0}{\partial y_i \partial y_j} + \frac{\partial \bar{H}_1}{\partial y_j} = 0, \qquad \frac{\partial \bar{H}_1}{\partial y_j} \neq 0$$

(which can be done assuming

$$\left|\frac{\partial^2 H_0}{\partial y_i \partial y_j}\right| \not\equiv 0),$$

giving a finally transformed Hamiltonian

$$H^* = H_0 + \kappa(\varepsilon)\left(\sum \xi_i \frac{\partial H_0}{\partial y_i} + \bar{H}_1\right) + \kappa^2 H_2^* \equiv H_0^* + \kappa^2 H_2^*.$$

From the way the ξ_i were determined we see that the frequencies

$$\frac{\partial H_0^*}{\partial y_i} = \frac{\partial H_0}{\partial y_i}.$$

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Subsequent transformations can be effected in the same way, eliminating effects due to additional secular terms and maintaining the initial frequencies for this solution.

4.2. Deep resonance

It is common in problems of resonance to follow Bohlin's suggestion (1889) and expand the Hamiltonian in powers of the square root of the small parameter (see, for example, Garfinkel, 1966; Jupp, 1973). Thus a linear elimination method like the von Zeipel technique would eliminate, successively, periodic terms multiplied by a single such power at each step. Successful application of the technique described here, in which the perturbation is eliminated to continually increasing powers of this parameter at each step, would be especially valuable in such problems.

5. Modifications of the Method

Since the initial formulation of this technique, it has been modified to make the generating function at each step satisfy the determining Equation (III) only to the order of the perturbation at that step, rather than identically as described. This procedure results in a significant reduction in the number of inter-term multiplications required.

Although the present method applies only to Hamiltonian systems, it could be applied to general systems Hamiltonized utilizing Kamel's procedure (1971). Alternatively, the method could be modified to be directly applicable to such non-Hamiltonian systems in much the same way as Henrard (1970) and Kamel (1970) generalized the linearly eliminating canonical algorithms respectively of Deprit (1969) and Kamel (1969). Research is currently being carried out in this direction.

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